THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 04

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** aqueous solution	Dissolved in water
2	**(d)** $Mg(OH)_2$	A mild base
3	**(b)** Incorrect — when it is acid rain its pH is less than 5.6	Dissolved gases lower pH
4	**(a)** diamond	Strong 3-D network
5	**(c)** magnesium hydroxide and hydrogen	$Mg + 2\,H_2O \longrightarrow Mg(OH)_2 + H_2$
6	**(a)** C_6H_{12}	Saturated ring
7	**(c)** sunlight	Photochemical
8	**(a)** Correct	Low pressure
9	**(d)** axon	Carries impulse away
10	**(a)** 2 to 8 days	Monthly cycle
11	**(a)** seminal vesicles and prostate	Semen
12	**(c)** 3 : 1	Phenotypic
13	**(b)** tt	Recessive needs two
14	**(c)** $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	Mirror formula has +
15	**(a)** blue/violet	Short wavelength
16	**(b)** four times	I^2
17	**(c)** nature of material and temperature	Not dimensions
18	**(b)** a sudden large current	Very low resistance
19	**(d)** freeze CFC production at 1986 levels	Ozone protection
20	**(d)** eagle	Top carnivore

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	(i) $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ (ii) $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$	1+1
22	NaCl is formed from a strong acid and a strong base, so it is neutral (pH 7).	2
23	Glucose; amino acids; fatty acids + glycerol	1+1
24	Organs (muscles/glands) that respond to impulses; e.g. arm muscles	2
25	Rose, jasmine, mango, orange (any two)	1+1
26	They inherit genes from both parents in new combinations.	2
27	$f = 1/4$ m = 25 cm; convex lens	1+1
28	False — it carries current only when there is leakage.	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	One substance oxidised and another reduced. (i) oxidising agent Fe_2O_3, reducing agent Al (ii) oxidising agent SO_2, reducing agent H_2S	1+1+1
30	(i) ionic (ii) XY_2 i.e. $MgCl_2$ (iii) yes	1+1+1
31	Phototropism: response to light — shoots bend towards light; geotropism: response to gravity — roots grow downward	1½+1½
32	Wind (maize), insects (sunflower), water (vallisneria), birds	3
33	Sunrise/sunset: light travels longer through atmosphere — blue scattered away, red reaches; noon: shorter path, little scattering	3

34	(i) increases (ii) reverses (iii) becomes zero	1+1+1
35	H_3O^+ decreases; pH increases; the base neutralises the acid and finally OH^- is in excess.	1+1+1
	OR: Cooking/preserving food; raw material for NaOH, baking soda, washing soda, bleaching powder; de-icing roads	1+1+1
36	Grass 10 000 J → rabbit 1000 J → fox 100 J; 90% lost as heat and used in life processes at each level	3
	OR: Food chain: single linear sequence, one organism per level, unstable; food web: interconnected chains, many alternatives, more stable	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) (a) beyond 2F, real, inverted, magnified (b) same side, virtual, erect, magnified (ii) $v = 20$ cm, $m = -1$, $h' = -3$ cm	3+2
	OR: (i) $P = 1/f$ in m, unit dioptre (ii) $P = 4 - 10 = -6$ D (iii) no, diverging	1+3+1
38	(i) H_2: single bond; O_2: double bond; N_2: triple bond — electron sharing (ii) weak intermolecular forces	3+2
	OR: (i) $2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$; $C_2H_5OH \xrightarrow{\text{conc}\cdot H_2SO_4} C_2H_4 + H_2O$ (ii) affects nervous system, liver (iii) blended with petrol (gasohol); clean fuel	2+2+1
39	(i) high resistance tungsten heats to high temperature and emits light (ii) to prolong filament life (iii) $H = Pt = 60 \times 300 = 18\,000$ J	2+1+2
	OR: (i) B (ii) $R \propto l/A$: $(2/0.5)/(1/1) = 4$ (iii) resistance of a unit cube, ρ (iv) same material and temperature (v) silver/copper	1+1+1+1+1
40	(i) anaerobic respiration (fermentation) (ii) ethanol, CO_2, energy (iii) cytoplasm (iv) muscles produce lactic acid (v) aerobic respiration	1+1+1+1+1
	OR: (i) exhaled air tube (ii) exhaled air has more CO_2 (iii) CO_2 (iv) exhaled air: more CO_2, less O_2 (v) respiration	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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